AI6124 · GROUP 18 · CONSULTATION NOTE

DENFIS against XGBoost and LSTM: what we tried, and what we should do next

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Task **Next-day DBS (`D05.SI`) log-return**

The nine-fold comparison is complete. DENFIS does not beat the baselines on forecast error, and we have since tested six improvement routes on validation data alone, all with negligible or no effect. We would like your guidance on two things: how "better than the original baselines" should be judged, and, if forecast error is the deciding measure, which direction you would like us to take.

01

What we built

Thirteen causal technical features, standardised on training data only, feed an ECM clustering stage that grows and merges rules online. Gaussian activations over those rules produce a first-order Takagi–Sugeno prediction, with consequents estimated by membership-weighted recursive least squares. Two variants are reported: **frozen**, fitted once on the training window, and **online**, which predicts first and then updates on the realised label.

Evaluation uses nine expanding walk-forward folds covering 2017–2025. Hyper-parameters are chosen on the preceding validation year only. The 2026 holdout is untouched and reserved for the scheduled freeze.

02

Current nine-fold results

Pooled over the nine test years, with 10 bp one-way trading cost.

MODEL	RMSE	R^2	DIRECTION ACC.	SHARPE	MAX DRAWDOWN	EXPOSURE
DENFIS frozen	0.012066	−0.019	0.501	**0.654**	**20.3%**	34.4%
XGBoost	**0.012009**	**−0.010**	**0.508**	0.238	32.2%	53.6%
LSTM	0.012043	−0.016	0.485	0.407	27.5%	30.7%
Buy and hold	—	—	—	1.112	38.6%	100%

Diebold–Mariano is not significant for any pair (DENFIS vs XGBoost $p = 0.219$; online vs frozen $p = 0.274$). The 95% stationary-bootstrap interval for the DENFIS Sharpe is [−0.041, 1.309], which spans zero. Every pooled R^2 is negative.

So there is no accuracy advantage, and no statistically established trading advantage. DENFIS has the highest point-estimate Sharpe and the lowest drawdown of the three models, but it sits below buy-and-hold and its interval spans zero.

03

The gap is not a test-set accident

On the validation years, before any test data is seen, DENFIS is already behind XGBoost in **seven of nine folds**. The shortfall is a property of the model on this data, not an unlucky test period. That is why we stopped trying to fix it by re-running the final evaluation.

04

Six improvement routes, all tested

Every route was evaluated on validation years only; no test year took part in any choice. The two rows marked exploratory were computed after the decision, as a check.

ROUTE	WHAT WE CHANGED	VALIDATION RESULT	DECISION
1. Hyper-parameter grid	Extended cluster radius `Dthr` from 1.5 to 2.0 and 3.0	Validation error fell only where the model collapsed to a single rule. At `Dthr = 3.0` three folds end with one rule, so the evolving structure disappears. Exploratory test check: pooled RMSE 0.012055 (−0.09%), DM p = 0.216.	Rejected
2. Feature set	Added three STI features: index return, index 20-day volatility, DBS–STI 20-day correlation	DENFIS improved in only 3 of 9 folds, median **+0.06%** (slightly worse). LSTM improved in 6 of 9 (mean −0.16%), XGBoost in 5 of 9.	Rejected
3. Inference rule	Kept only the *m* most activated rules, the original DENFIS mechanism; our implementation normalises over all rules	5 of 9 folds improved, mean −0.074%, median −0.066%.	Too small
4. Consequent learning	Robust RLS: correntropy at three scales, and Huber	Best-point mean −0.41%, driven by two folds. Across the full grid one was a single lucky point (7 of 20 points better, median **+7.2%**). The other was real but unstable (16 of 20 points, median −1.7%, some configurations blow up).	Rejected
5. Bagging the model	Eight bootstrap replicates, averaged predictions	At the point selection would actually pick: 3 of 9 folds win, mean **+0.07%**. Across the whole grid 12–19 of 20 points improve (median −2% to −19%): bagging repairs unstable configurations, but validation-based selection already avoids them.	Rejected
6. Blending with the baseline	50/50 blend of DENFIS and XGBoost, plus the best per-fold weight	50/50 beats XGBoost in 3 of 9 folds (mean +0.07%). Even with the weight fitted on the same validation data, **five of nine folds choose w = 0**, dropping DENFIS entirely.	Rejected

Routes 5 and 6 say the same thing from different angles: the model's variance is already controlled by validation-based selection, and its errors are too correlated with XGBoost's to be worth blending.

05

What the literature suggests, and what it does not

Published improvements to this model family are almost always compared against other fuzzy models, such as eTS, SAFIS, PANFIS or FLEXFIS, and rarely against a tuned gradient-boosting baseline. We could not find a study reporting an evolving fuzzy system beating a well-tuned GBM on daily equity returns.

The mechanisms the literature credits (type-2 fuzzy sets, participatory learning, kernel RLS, concept-drift detection, rule pruning, online bagging) mostly add capacity or adaptation. That is the opposite of what our diagnostics point to: on this data every capacity increase has made validation error worse.

06

What we deliberately did not do

We did not select parameters, features, blend weights or folds using test-year performance. Six exploratory directions could each have been reported as a win by quoting the best fold or the best parameter point. Each is reported above with the grid-wide result that contradicts it. A fresh run reproduces the committed results to about 1e-13.

07

Questions for you

1. When the brief says the improvement must be "better than the original baselines", which measure decides it: forecast error such as RMSE or directional accuracy, risk-adjusted trading performance such as Sharpe and drawdown, or an overall judgement?
2. Under that definition, does our current position qualify? DENFIS is worse on forecast error (0.012066 against 0.012009), but better than both baselines on Sharpe (0.654 against 0.238 and 0.407) and on drawdown (20.3% against 32.2% and 27.5%), with every statistical test non-significant.
3. If forecast accuracy is the deciding measure, is an honest negative result acceptable when the comparison is rigorous and every alternative has been tested and documented? We would rather report this than tune on the test set.

4 If a positive result is required, which of these should we pursue before the 15 November freeze?

(a) Change the prediction target, for example volatility or a multi-day horizon, where local fuzzy rules have a structural reason to help. This changes the task, so we would not do it without your agreement.

(b) Report a DENFIS plus XGBoost blend as a separate "system" result while keeping DENFIS as the main model. Our validation checks show it beats neither on accuracy, but it is the most defensible framing of a combined model.

(c) Keep the current protocol and report the negative result as the finding, with the evidence in section 04 as the analysis.

08

What we would do under each answer

YOUR ANSWER	OUR NEXT STEP	TIME NEEDED
Risk-adjusted performance counts	Freeze as planned on 15 November. Main table unchanged, trading metrics promoted to the headline.	None
An honest negative result is acceptable	Same, with section 04 written up as the analysis section.	None
A positive accuracy result is required	Protocol v1.2 with a changed target or an ensemble as the main model. Both need a full re-run and your sign-off.	2–3 days
None of these	We would like your guidance on direction before the 15 November freeze.	—

Supporting artifacts: `results/summary.csv`, `results/statistical_tests.json`, `results/analysis/validation_grid.csv`, `docs/results_table.md`, `docs/statistical_tests.md`, `docs/leakage_audit.md`. Final notebook: `writing/AI6124_Project_Group18_HuJunyu_LinJingzhao.ipynb`.